NOTE

The greater the value of β_2 , the more peaked the distribution.

(i) The normal curve and other curves with $\beta_2 = 3$ are called mesokurtic.

(ii) When the value of β_2 is greater than 3, the curve is more peaked than the normal curve, i.e., leptokurtic.

(iii) When the value of β_2 is less than 3 the curve is less peaked than the normal curve, i.e., platykurtic.

Example 6.19

Find the value of β_2 for the following data $\mu_1 = 0\ \mu_2 = 4\ \mu_3 = 0\ \mu_4 = 37.6$

Solution:

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\beta_2 = \frac{37.6}{4^2} = \frac{37.6}{16} = 2.35 < 3$$

$\beta_2 < 3$, The curve is platykurtic.

6.8 Box plot

A box plot can be used to graphically represent the data set. These plots involve five specific values:

(i) The lowest value of the data set (i.e., minimum), (ii) Q_1 (iii) The median (iv) Q_3, (v) The highest value of the data set (i.e maximum)

These values are called a five- number summary of the data set.

A box plot is a graph of a data set obtained by drawing a horizontal line from the minimum data value to Q_1 and a horizontal line from Q_3 to the maximum data value, and drawing a box by vertical lines passing through Q_1 and Q_3, with a vertical line inside the box passing through the median or Q_2.

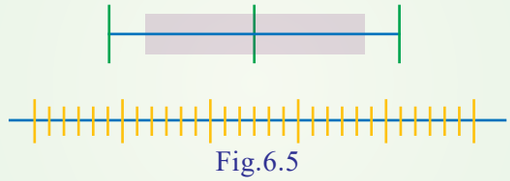


Fig.6.5

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